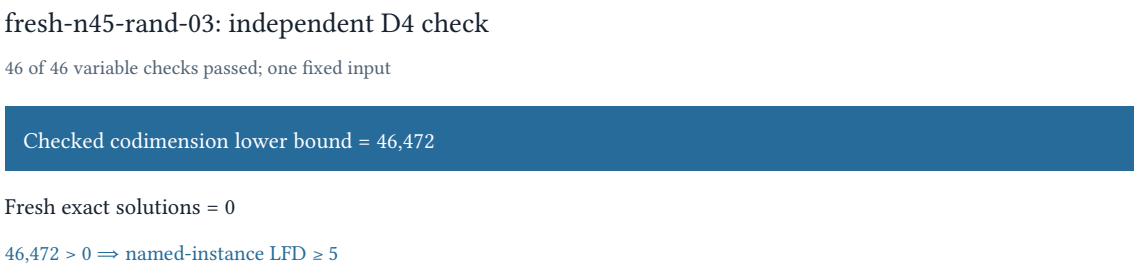
Certified n45 random03: LFD at least five

The independent D4 checker accepted all 46 variable multiplication checks for fresh-n45-rand-03. It certified codimension at least **46,472** and fresh exhaustive enumeration found **0** solutions. Thus the strict inequality establishes **LFD at least 5** for this named Boolean presentation.

The certificate is 1,045,099,360 bytes with SHA-256 3e023fd028e1e301aabaf1723ac2f1d084b07949bafd4c05a9f74c5dfaac7188. The collector checked all 47 worker artifacts and successful terminal return; three controls passed. Public certificate and evidence readback were verified.



Admission passed after 0.008 seconds at 29,541,203,968 bytes available. Producer and checker took 2256.234 and 491.859 seconds. The original panel now has 31 terminal, 24 certified LFD-at-least-five, and one remaining input. Frozen source a4cc5210cce0e83bbd8faf54b6186a97c31241c5; protocol a24072c1a350f428727d979b13fd9e9f24db2a31b1daf7787bcfe12a61cff8f8.

Input n=45, k=23, seed 13651200446574642926, degree 4. Runtime command and paths are in report.md and the frozen plan. CPU time was not recorded.